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Ecological inference in electoral contexts

Theoretical foundations, methodological developments and applications

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Abstract

This thesis investigates the problem of ecological inference, which arises when the object of interest is the relationship between variables at the individual level, but data are only available at the aggregate level.

Section 1 presents the problem of ecological inference and its relevance in various contexts.

Section 2 introduces general notation and definitions for the problem of ecological inference, and possible assumptions that can be made to identify the individual-level relationships from aggregate data, with a particular focus on CAR assumptions and the method proposed by McCartan and Kuriwaki (2026a).

Section 3 presents an adaption of the model for the specific context of electoral analysis.

Section 4 presents a synthetic example to illustrate the method and its performance in a controlled setting, where the true individual-level relationships are known.

Section 5 presents an application of the method to 2020 US presidential elections in Texas.

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1 Introduction

Ecological inference fallacy is the problem that arises when we are interested in relationships between variables at the individual level but we only observe data at the aggregate level.

A famous example of that problem is the one presented by Robinson (1950) for the relationship between illiteracy and race, and illiteracy and migration in the United States. In that case, the author observed that areas with a higher proportion of Black population, as well as areas with a higher proportion of migrants, had lower illiteracy rates. However, at the individual level, Black people and migrants were more likely to be illiterate than White and non-migrant people. The explanation for this apparent paradox is that Black and migrant people were more likely to live in or move to areas with lower illiteracy rates, and therefore the aggregate data masked the individual-level relationship.

We can also think of economic examples.

Consider, for instance, a retail store chain that wants to investigate the consumption habits of young people by observing sales data from different stores. The retail store chain might know the quantities of goods sold in different stores and, at the same time, demographic data about consumers in the different areas served by those stores. Therefore, it would be tempting to link the percentage of young people to the quantities of different goods sold, suggesting that goods that sell more in stores with younger consumers are purchased by young consumers.

However, that could lead to false findings. In fact, when we consider data on cities, it might well be that stores with a greater share of young consumers are located in central areas near universities, where richer people also usually live. Therefore, even if young people might be more price-sensitive consumers, their habits would be masked by their neighbors' habits.

In any case, the most obvious application of ecological inference concerns electoral behavior. We are interested in the relationship between individual-level characteristics and voting behavior, but, obviously (and fortunately!), we cannot observe how individuals vote. We are therefore unable to link voting behavior directly to individual-level characteristics. In these cases, we are forced to rely on aggregate data, ideally at the most granular level of aggregation available. However, even at the most granular level of aggregation, individual-level relationships may be severely masked.

This issue is particularly relevant for political economists, who often study the relationships between voters' characteristics, electoral outcomes, and public policies.

Kuriwaki and McCartan (2026) present a particularly convincing example of this problem using the 1968 US presidential election. In addition to the Republican and Democratic candidates, Richard Nixon and Hubert Humphrey, respectively, there was a relevant third-party candidate, George Wallace, who led the segregationist American Independent Party. George Wallace performed best in the Southern states, where the proportion of the Black population was generally higher than in the rest of the country. Therefore, a naive analysis of the aggregate data might suggest that Black people were more likely to vote for George Wallace, since states with larger Black populations also had higher proportions of votes for him. That would, of course, be a false conclusion: Black people certainly did not favor George Wallace, whose platform was segregationist and overtly racist.

Kuriwaki and McCartan (2026) go on to show that, even at more granular levels of aggregation, such as the county level, the relationship between the proportion of the Black population and the proportion of votes for George Wallace remains positive when considering all counties in the nation. Moreover, even when restricting the analysis to specific states, such as North Carolina, we still observe a positive relationship: the higher the proportion of the Black population in a county, the higher the proportion of votes for George Wallace.

The key is that, while Black voters were relatively homogeneous in their voting behavior—generally favoring the Democratic candidate and certainly not voting for George Wallace—White voters were much more heterogeneous. In particular, White people with segregationist views were more likely to live in areas with a higher proportion of Black residents.

The same mechanism is at work in the retail-store example: while young consumers might be relatively homogeneous in their spending habits, the larger group of older consumers may have different habits and drive the aggregate relationship.

Robinson (1950), who raised the issue of ecological fallacy with the illiteracy example, came to the conclusion that ecological fallacy would constitute an inextricable problem and he concluded his paper telling that he was

> ... aware that this conclusion has serious consequences, and that its effect appears wholly negative because it throws serious doubt upon the validity of a number of important studies made in recent years. The purpose of this paper will have been accomplished, however, if it prevents the future computation of meaningless correlations and stimulates the study of similar problems with the use of meaningful correlations between the properties of individuals.

Econometrics in 1950, however, had considerably fewer tools than it does today, and we might therefore think that some of these techniques could be successfully deployed to address the ecological fallacy.

In particular, we might be tempted to argue that the problem could be solved by including covariates.

That, however, is not necessarily sufficient. In fact, when Kuriwaki and McCartan (2026) examine the results of presidential elections in North Carolina, they are implicitly including a state indicator and interacting it with the proportion of Black voters. Indeed, their estimate can be interpreted as the coefficient on the share of Black voters, conditional on the county being located in North Carolina.

Analogously, we might include every possible interaction between variables and functional form, or impose a nonlinear relationship between the share of Black voters and electoral outcomes. This could produce a positive relationship between the share of Black voters and segregationist votes at relatively low levels of Black population, which would reverse once Black voters became the majority.

Another method we might consider is the inclusion of fixed effects. However, fixed effects require repeated observations of the same unit, which are not available in our examples.

All these examples highlight the common problem with using traditional econometric techniques to address ecological inference: the discrepancy between the unit of observation and the quantity we seek to estimate.

Indeed, each of these techniques may help identify relationships between variables at the level of the unit of observation, but none would solve the ecological fallacy of composition.

Even our most naive estimates may correctly describe relationships at the level of the units of observation. It may be true that counties with more migrants have lower illiteracy rates, even after accounting for other variables or effects; that consumers in areas with younger populations spend more on expensive goods; or that precincts with higher shares of Black voters favor segregationist candidates.

What these estimates cannot tell us is whether migrants have lower or higher illiteracy rates, whether younger consumers spend more or less on expensive goods, or how Black people vote.

To answer these questions, we must introduce notation that explicitly accounts for the aggregate nature of our units of observation and impose a set of assumptions.

Only then can we move from the agnostic conclusion of Robinson (1950) to ecological estimates.

In the next section, I will introduce a general notation for ecological inference problems and show how different assumptions about the distribution of individual-level characteristics within geographical units of observation can lead to different ecological inference methods.

In the following section, I will present a more specific model for electoral analysis.

Both sections draw heavily on Kuriwaki and McCartan (2026), which reviews ecological inference methods, and on McCartan and Kuriwaki (2026a), which presents the semiparametric method for ecological inference implemented in the R package `seine` (McCartan and Kuriwaki (2026b)) and used in this dissertation.

The notation I present is nevertheless slightly different and better suited to describing the issues considered in this dissertation.

After presenting the model, I will use synthetic examples to illustrate how the method works and assess its performance in a controlled environment.

Finally, I will apply the method to real data from the 2020 US presidential election in Texas.

2 Notation for ecological inference

We consider a population of N individuals, denoted by $i \in \{1, \dots, N\}$.

Each individual i is characterized by a set of variables. Among them, we consider a response variable Y_i and a categorical variable X_i.

The response variable can be either quantitative or categorical, and will be denoted accordingly. In the quantitative case, we write $Y_i \in \mathbb{R}$. In the categorical case, we write $Y_i \in \{0, 1\}^P$, where P is the number of possible categories. That is, Y_i is a vector whose entries are all equal to 0, except for the entry corresponding to the category of individual i, which equals 1.

Thus, for example, $Y_i = (1, 0, 0, \dots, 0)$ if the response variable takes the first category, $Y_i = (0, 1, 0, \dots, 0)$ if it takes the second category, and so on.

The categorical variable X_i, instead, is denoted by $X_i \in \{0, 1\}^K$, where K is the number of possible categories. It encodes the category to which individual i belongs. We shall denote the categories by $k \in \{1, \dots, K\}$.

The total population is partitioned into G geographies, denoted by $g \in \{1, \dots, G\}$. Each individual i belongs to one and only one geography, which we denote by G_i.

Geographies are what we previously referred to as units of observation. This means that we are never able to observe Y_i and X_i directly; rather, we only observe variables at geography level.

For each geography g, the population is $N_g = \sum_i \{1 \mid G_i = g\}$.

Moreover, $\overline{Y}_g = \frac{1}{N_g} \sum_i \{Y_i | G_i = g\}$ and $\overline{X}_g = \frac{1}{N_g} \sum_i \{X_i | G_i = g\}$.

$\overline{Y}_g$ can be either a scalar, if the response variable is quantitative, or a vector of shares $\overline{Y}_g \in [0,1]^P$, if the response variable is categorical. In the latter case, each entry of the vector represents the share of the corresponding category, and the identity $\sum_{p=1}^{P} \overline{Y}_{gp} = 1$ follows from the categorical nature of the response variable.

$\overline{X}_g$ is a vector of shares $\overline{X}_g \in [0,1]^K$, given the categorical nature of X_i.

2.1 Global and local estimands

In general, we are interested in the average value of Y_i conditional on belonging to category $k \in \{1,\dots,K\}$.

We shall denote this quantity by $\beta_k = \frac{\sum_i \{Y_i | i \in I_k\}}{N_k}$, where I_k is the subset of people belonging to category k, and N_k is the cardinality of I_k.

As before, β_k is a scalar if the response variable is quantitative and a vector of shares of dimension P if the response variable is categorical.

When we consider all categories $k \in \{1,\dots,K\}$, we obtain either a vector $\beta \in \mathbb{R}^K$ or a matrix $\beta \in [0,1]^{P \times K}$.

We shall call β the global estimand.

Since we do not observe individuals, it is useful to define analogous quantities at the geography level.

The local estimand for category k and geography g is defined as $\beta_{gk} = \frac{\sum_i \{Y_i | i \in I_{gk}\}}{N_{gk}}$, where I_{gk} is the subset of people belonging to category k and geography g and N_{gk} is its cardinality.

Analogously, we can collect all β_{gk} corresponding to the same geography g but different categories k into β_g, a vector of dimension K or a matrix of dimension $P \times K$.

These definitions yield two accounting identities.

The first one links the global estimand to the local estimands: $\beta_k = \frac{\sum_{g=1}^{G} \beta_{gk} * N_{gk}}{N_k}$, i.e. the global estimand is the weighted average of local estimands, weighted by the population of each geography.

This identity suggests a strategy that starts from local estimands, which may also be of interest in their own right, to recover the global estimand.

The second accounting identity is at the geography level: $\overline{Y}_g = \beta_g' \overline{X}_g$.

One consequence of this identity is that each observation contributes P or $P \times K$ parameters, making the local problem unidentifiable.

2.2 Different assumptions, different strategies

2.2.1 Common Coefficient assumption

The simplest way to escape the identification problem is to assume that $\beta_g = \beta, \forall g$. Consequently, a simple linear regression can be adopted. Indeed, under this assumption, the second accounting identity becomes $\overline{Y}_g = \beta' \overline{X}_g$.

This assumption, however, is particularly stringent and can be easily falsified by the data by finding two geographies with $\overline{X}_g$ and $\overline{Y}_g$ that are incompatible with the same β.

A numerical example is the following.

Consider two geographies g_1 and g_2 with Y_i categorical and two categories (Yes and No), and X_i categorical with two categories (A and B).

Suppose that in geography g_1 we have 50% of individuals in category A and 50% in category B, while in geography g_2 we have 60% in category A and 40% in category B.

Suppose further that in geography g_1, 90% of individuals answered Yes and 10% answered No (denoted by $\overline{Y}_{g_1} = 0.9$), while in geography g_2, 10% answered Yes and 90% answered No (denoted by $\overline{Y}_{g_2} = 0.1$).

In this case, it is impossible to find a single β that satisfies both $\overline{Y}_{g_1} = \beta'\overline{X}_{g_1}$ and $\overline{Y}_{g_2} = \beta'\overline{X}_{g_2}$.

Indeed, to satisfy the first equation, we would need at least 80% of individuals in category B in geography g_1 to have answered Yes (if we impose $\beta_{g_1A} = 1$ and $\beta_{g_1B} = 0.8$). The first equation requires $\beta_{g_1B} \geq 0.8$.

However, this is incompatible with the second equation, which requires that at most 25% of individuals in category B in geography g_2 answered Yes (if we impose $\beta_{g_2A} = 0$ and $\beta_{g_2B} = 0.25$). The second equation requires $\beta_{g_2B} \leq 0.25$.

2.2.2 CCAR assumption

A slightly more relaxed assumption is allowing for variations in β_g, while imposing that the variations in β_g are independent of $\overline{X}_g$: $E[\beta_g|\overline{X}_g] = \beta, \forall g$; or, equivalently, $E[(\beta_g - \beta)|\overline{X}_g] = 0, \forall g$.

That assumption, which cannot be falsified by observable data per se, would still allow us to use linear regression.

Indeed, treating geographies g as units of observation and considering $\overline{Y}_g$ and $\overline{X}_g$, the model is equivalent to a standard linear model of the form

$$\overline{Y}_g = \beta_g'\overline{X}_g = \beta'\overline{X}_g + \epsilon_g, \qquad \epsilon_g \equiv (\beta_g - \beta)'\overline{X}_g.$$

Under the assumption that $E[(\beta_g - \beta) \mid \overline{X}_g] = 0$, we have

$$E[\epsilon_g \mid \overline{X}_g] = E[(\beta_g - \beta)'\overline{X}_g \mid \overline{X}_g] = E[(\beta_g - \beta) \mid \overline{X}_g]'\overline{X}_g = 0,$$

so linear regression can be used.

As emphasized by Kuriwaki and McCartan (2026), this is essentially what Goodman (1953) described, albeit with a more primitive notation, when discussing conditions under which linear regressions could be used in ecological inference, thereby overcoming Robinson (1950)'s concerns.

This assumption is called Coarsening Completely At Random (CCAR), because it is equivalent to assuming that the way in which individuals are divided into geographies is independent of other characteristics of the geography: literate and illiterate White people, more price-sensitive and less price-sensitive older consumers, and segregationist and non-segregationist White voters are randomly assigned to counties, retail stores, and electoral precincts.

Equivalently, CCAR would allow the following identification:

$$\beta = E[\overline{X}\overline{X}']^{-1}E[\overline{X}'\overline{Y}].$$

In any case, when performing ecological regression, it is rarely the case that this assumption holds: in the examples discussed in the Introduction, it would imply that White and non-migrant individuals share the same average illiteracy rates (β_g), regardless of the percentage of Black and migrant individuals in their county ($\overline{X}_g$); that older consumers share the same spending habits (β_g), regardless of the share of young people in their area ($\overline{X}_g$); and that White voters share the same voting behavior (β_g), regardless of the percentage of Black voters in their precinct ($\overline{X}_g$).

Kuriwaki and McCartan (2026) express this point as follows:

> Is coarsening completely at random plausible in practice? Consider again the 1968 election example. In this case, CCAR means that the preference of White voters for Wallace is unrelated to the proportion of Black voters in a county or the population of the county. For instance, this means that all of the following groups support Wallace to the same extent in North Carolina in expectation:

White voters in rural lowland counties with a higher proportion of Black voters, White voters in urban counties, and White voters in mountainous counties with few racial minorities. CCAR is implausible in this setting, and is incompatible with the social scientist's interest in how different individuals sort into different coarsened geographies.

2.2.3 Adding covariates: CAR assumption

So far, we have not discussed the possibility of adding covariates to the model, which would allow us to relax the CCAR assumption.

We now introduce a set of covariates Z_g at the geography level. Since geographies are our units of observation, these covariates may either be aggregates of individual characteristics or characteristics of the geography itself. This distinction is not relevant here, and we will later provide examples of both types. A consequence for notation is that we denote covariates as Z_g without an overline, because they are not necessarily averages of individual characteristics.

Thanks to covariates, we can relax the CCAR assumption and instead consider the following condition:

$$E[\beta_g \mid \overline{X}_g, Z_g, N_g] = E[\beta_g \mid Z_g].$$

This assumption is called Coarsening At Random (CAR). It is equivalent to assuming that the way in which individuals are divided into geographies is independent of other characteristics of the geography, conditional on covariates: once we account for adequate covariates, literate and illiterate White people, more price-sensitive and less price-sensitive older consumers, and segregationist and non-segregationist White voters are randomly assigned to counties, retail stores, and electoral precincts.

As a consequence, geographies may have different β_g, and, contrary to CCAR,

$\overline{X}_g$ can affect β_g. However, we assume that there exist covariates Z_g that capture the effect of $\overline{X}_g$ on β_g.

Therefore, we can write β_g as a function of the covariates:

$$\beta_g = E[\beta_g \mid \overline{X}_g, Z_g, N_g] + \zeta_g = E[\beta_g \mid Z_g] + \zeta_g, \qquad E[\zeta_g \mid \overline{X}_g] = 0.$$

Without loss of generality, we can express $E[\beta_g \mid Z_g]$ as a vector-valued function:

$$E[\beta_g \mid Z_g] = f(Z_g).$$

Then, using the second accounting identity, we obtain a model linking $\overline{Y}_g$ to $\overline{X}_g$ and Z_g:

$$\overline{Y}_g = \beta_g' \overline{X}_g = f(Z_g)' \overline{X}_g + \zeta_g' \overline{X}_g.$$

Taking conditional expectations on both sides, we obtain

$$E[\overline{Y}_g \mid \overline{X}_g, Z_g, N_g] = f(Z_g)' \overline{X}_g.$$

This makes the problem identifiable: instead of estimating a different set of parameters β_g for each observation, as suggested by the second accounting identity, we estimate a single functional form $f(Z_g)$ that links geographical covariates to the unobserved β_g.

The simplest estimation strategy would be to impose a linear functional form for $f(Z_g)$: $f(Z_g) = \gamma' Z_g$. This would make the whole model linear and therefore estimable by linear regression:

$$E[\overline{Y}_g \mid \overline{X}_g, Z_g, N_g] = f(Z_g)' \overline{X}_g = (\gamma' Z_g)' \overline{X}_g = Z_g' \gamma \overline{X}_g.$$

As discussed by McCartan and Kuriwaki (2026a), this condition is stringent but not necessary.

In fact, McCartan and Kuriwaki (2026a)'s methodology is a form of semi-parametric estimation that uses the linear structure of the second accounting identity, $\overline{Y}_g = \beta_g' \overline{X}_g$, while allowing for a nonparametric form of $f(Z_g)$. It requires only two additional assumptions.

These assumptions are referred to as positivity and bounded N.

The first additional assumption requires sufficient variation in $\overline{X}_g$ after controlling for Z_g, and it is analogous to the overlap assumption in causal inference: for each value of Z_g, there must be a non-zero probability of observing each value of $\overline{X}_g$.

Without this assumption, it would be impossible to disentangle the effect of $\overline{X}_g$ on $\overline{Y}_g$ from the effect of Z_g on β_g, and the problem would remain unidentified.

The second additional assumption requires that there is no overwhelmingly large geography whose effect dominates that of other geographies.

In the next section, I will present the model for electoral competitions and discuss how it satisfies these assumptions.

3 A model for electoral competitions

Electoral competitions can be viewed as situations in which each voter i chooses among P possible candidates, so that the response variable Y_i is a P-dimensional vector with $Y_i \in \{0,1\}^P$.

However, in both the synthetic example and the application to U.S. electoral data, I will treat Y_i as a binary choice. In the synthetic example, this is a consequence of the referendum rule, where only two options are available (vote Yes or vote No). In the empirical application, it follows from the American two-party system, where two parties dominate the political landscape and the remaining parties are not relevant for the analysis.

Hence, we can focus on one of the two possible votes and denote Y_i by a binary indicator $Y_i \in \{0,1\}$, which takes value 1 if the vote is cast in favor of the reference choice and 0 otherwise.

Electoral precincts are the geographical units in which votes are aggregated, and they are the units of observation in the analysis. Therefore, we denote by g a generic electoral precinct and by G the number of electoral precincts.

As a consequence, the observed variable $\overline{Y}_g$ is a scalar with $\overline{Y}_g \in [0,1]$ that represents the share of the reference choice in precinct g.

We then consider a partition of individuals into K mutually exclusive and exhaustive categories, which are the categories of the variable $X_i \in \{0,1\}^K$. Examples of such partitions could be the partition of voters into mutually exclusive and exhaustive income classes, or the partition of voters into mutually exclusive and exhaustive ethnic groups.

Finally, each electoral precinct g is characterized by a vector of covariates

$Z_g \in \mathbb{R}^d$, which may either describe geographic features (e.g., the location of the precinct in a particular region) or arise as aggregates of individual-level covariates (e.g., the education level of the voters in that precinct).

It is now necessary to introduce the assumptions that allow us to identify the model and estimate the parameters of interest.

The positivity assumption depends on the choice of covariates Z_g and on the absence of collinearity between them and $\overline{X}_g$, so that the model is identifiable.

The bounded N assumption can be verified by verifying that no electoral precincts are too large.

Finally, we can impose a Coarsening At Random (CAR) assumption through an individual-level condition, which, following McCartan and Kuriwaki (2026a), we may call CAR-IND.

CAR-IND states that the individual-level response variable Y_i is independent of the geography g, conditional on the individual-level covariates X_i and the geography-level covariates Z_g. Formally, this can be expressed as follows:

$$E[Y_i \mid G_i = g, X_i = x_i, Z_{G_i} = z_g] = E[Y_i \mid X_i = x_i, Z_{G_i} = z_g].$$

From a practical point of view, this assumption implies that the covariates capture all the effects of geography on the individual-level response variable Y_i. In the electoral example, this means that White voters in precinct g with characteristics Z_g have the same probability of voting for the segregationist candidate as White voters in a different precinct g' with the same characteristics $Z_{g'} = Z_g$.

Moreover, because $f(Z_g)$ is not necessarily injective, it is possible that two different electoral precincts g and g', even with different characteristics $Z_g \neq Z_{g'}$, have the same value $f(Z_g) = f(Z_{g'})$, and therefore the same expected value $\beta_g = \beta_{g'}$.

McCartan and Kuriwaki (2026a) show that CAR-IND implies CAR.

Therefore, to perform semiparametric estimation of the model, we need to check bounded N and then select covariates Z_g that satisfy the testable positivity assumption and the untestable CAR-IND assumption.

4 A synthetic example

In this section I will present a synthetic example that helps explaining why the problem is relevant and shows the difference performance between naive OLS and the semiparametric estimation proposed by McCartan and Kuriwaki (2026a) and embedded in R package `seine` (McCartan and Kuriwaki (2026b)).

In the synthetic example we are the ones who generate the data and therefore we know the exact model of the reality. I will show that, estimating the quantities of interest with different methods, we obtain different results and OLS provides biased estimates.

4.1 Italian constitutional referendum: a motivating example

We can exploit the constitutional referendum in Italy in March 2026 to show a possible way in which the assumptions behind OLS are violated, and therefore we can generate a dataset that encompasses the problem.

Let us suppose that we want to estimate how did voters with different levels of income vote.

If we look at data at regional level, we would be tempted to say that richest people voted more in favour of the government proposal. In fact, the only three regions where the votes in favour were more than the votes in contrast (Lombardy, Veneto and Friuli-Venezia Giulia) are among the richest regions in Italy. On the contrary, the regions were No got more than 60% are among the poorest regions of Italy (Basilicata, Campania, Sicily).

However, once we look at data at municipal level, we find that, within the regions, the relationship is nearly inverse: No votes were more frequent in urban areas and municipalities where income is relatively higher.

That divergence makes us think that if we were able to observe results at individual level, we could find even different relationships between income and voting behavior.

Therefore, we can now generate a synthetic dataset that presents a structure that resembles some stylized facts that we observe in the constitutional referendum results and other notions about electoral behaviors in Italy.

4.2 The synthetic experiment

The synthetic experiment is constructed as follows.

We assume that election takes place in a Country which has three different regions: North, Centre and South.

Each region is divided in 100 precincts, so that in the end we observe 300 precincts: 100 precincts in the North region, 100 in the Centre region, 100 in the South region.

Each precinct has the same voting population N_g, which is then normalised to 1.

For each electoral precinct, we observe the percentage of high-income and low-income voters ($\overline{X}_g$) and the electoral results ($\overline{Y}_g$).

Within each income category, there is high variability in their electoral behaviour across different precincts.

In particular, we impose that high-income people in the North and South precincts have a high propensity of voting in favour of the reform, while in the Centre this percentage is much lower. As for low-income people, we impose that in North they have high propensity of voting in favour, while in the Centre and in the South this propensity is much lower.

At the same time, we impose that the percentage of high-income voters is highest in the North and lowest in the South and intermediate in the Centre.

4.2.1 Data generating process

For each region $r \in \{\text{North}, \text{Centre}, \text{South}\}$ and for each of the 100 precincts in that region, we draw the share of high-income voters from a truncated normal distribution,

$$\overline{X}_{gr}^{H} \sim \text{TruncNormal}(a = 0, b = 1, \mu_r, \sigma = 0.2),$$

with means $\mu_{\text{North}} = 0.7$, $\mu_{\text{Centre}} = 0.5$, and $\mu_{\text{South}} = 0.3$. The low-income share is then set to $\overline{X}_{gr}^{L} = 1 - \overline{X}_{gr}^{H}$. Hence, on average, Northern precincts have higher shares of high-income voters, Southern precincts have lower shares, and Central precincts lie in between. The use of Truncated Normal distribution allows to have quite high variability within geography ($\sigma = 0.2$), but without having negative shares or shares above 1.

Subsequently, I assign to each precinct a propensity to vote in favour of the reform within each income group and I encode it in a vector $\beta_g = \begin{pmatrix} \beta_g^H \\ \beta_g^L \end{pmatrix}$.

β_g^H and β_g^L are extracted again from truncated normal distributions with different means depending on the region.

$$\beta_{gr}^{H} \sim \text{TruncNormal}(a = 0, b = 1, \mu = \eta_r^H, \sigma = 0.05),$$
$$\beta_{gr}^{L} \sim \text{TruncNormal}(a = 0, b = 1, \mu = \eta_r^L, \sigma = 0.05).$$

Means η_r^H and η_r^L are set as follows:

$$\eta^H_{\text{North}} = 0.6, \quad \eta^L_{\text{North}} = 0.7,$$
$$\eta^H_{\text{Centre}} = 0.3, \quad \eta^L_{\text{Centre}} = 0.3,$$
$$\eta^H_{\text{South}} = 0.6, \quad \eta^L_{\text{South}} = 0.2.$$

The key feature of this design is that the income-specific propensities are heterogeneous within each group and also vary across regions. High-income voters in the North and South have a strong tendency to support the reform, while in the Centre the same group is much less favourable. Low-income voters have a relatively high propensity to support the reform in the North, but not in the Centre or South.

Finally, the observed precinct-level outcome ($\overline{Y}_g$, i.e., the share of voters in favour of the reform in precinct g) is generated as the weighted average of these two group-specific propensities:

$$\overline{Y}_g = \overline{X}_g^H \beta_g^H + \overline{X}_g^L \beta_g^L.$$

In the end, we have a dataset that for each electoral precinct g contains the following information: the share of high-income voters $\overline{X}_g^H$, the share of low-income voters $\overline{X}_g^L$, the observed outcome $\overline{Y}_g$, the location of the precinct (North, Centre or South), the expected propensity to vote in favour of the reform for high-income voters η_r^H and for low-income voters η_r^L (equal across geographies of the same region), and the actual propensities β_g^H and β_g^L (specific of each geography).

4.2.2 From DGP to the model

The generated dataset strictly satisfies the assumptions stated in previous sections.

The response variable Y_i is a binary variable $Y_i \in \{0,1\}$ and it is observed only at geography level as a share $\overline{Y}_g \in [0,1]$.

Population is partitioned across geographies and, within each geography, is partitioned into two groups (high-income and low-income voters), so that for each geography g we observe $\overline{X}_g = \begin{pmatrix} \overline{X}_g^H \\ \overline{X}_g^L \end{pmatrix}$ with $\overline{X}_g^H + \overline{X}_g^L = 1$.

Moreover, we can interpret the region of each precinct as a covariate $Z_g = \{0,1\}^3$, which in this case does not arise from individual characteristics but rather from the geographical location.

$\overline{Y}_g$, $\overline{X}_g$ and Z_g are observed, while β_g (and their corresponding η_r) are unobserved and are the goals of the estimation.

Bounded N is satisfied as N_g is equal across different geographies.

Positivity is satisfied as we have allowed for variability in the shares of high-income and low-income voters across geographies within each region.

Finally, CAR-IND is almost completely satisfied, thanks to the fact that β_g are generated from a distribution with low variance ($\sigma = 0.05$) that depends on the covariate Z_g.

In fact, a complete adherence to CAR-IND would require β_g to be equal across geographies of the same region, but I have allowed for some variability in order to make the example more realistic and more responding to the expected ability of the covariates to explain the variation in the propensitities.

In our case, $E[Y_i|G_i = g, X_i = x_i, Z_{G_i} = Z_g] = \beta_g$ is just slightly different from $E[Y_i|X_i = x_i, Z_{G_i} = Z_g] = \eta_r$.

The independence assumption can be stated also at aggregate level, which in fact is the identifying assumption of the semiparamtric estimation, without explicit dependence on CAR-IND.

$$E[\beta_g|\overline{X}_g, Z_g, N_g] = E[\beta_g|Z_g]$$

Although we have allowed for some variability in β_g across geographies of the same region, the expected value of β_g is equal to η_r and therefore it depends only on Z_g.

4.3 Comparison of results

As previously mentioned, in our dataset we know the true values of β_g for each geography g, therefore we are able to obtain the true values of the estimands of interest β^H and β^L by aggregating the true values of β_g as stated in the first accounting identity in Section 2: $\beta^H = \frac{\sum_{g=1}^{G} \beta_g^H * N_g^H}{\sum_g N_g^H}$ and $\beta^L = \frac{\sum_{g=1}^{G} \beta_g * N_g^L}{\sum_g N_g^L}$.

In our case, we have that $\beta^H = 0.503$ and $\beta^L = 0.343$.

high	low
0.503	0.342

Tab. 1: True coefficients up to approximation

4.3.1 OLS estimates

The naive OLS specification, implicitly assuming CCAR, would rely on the following regression: $\overline{Y_g} = \beta^H * \overline{X}_g^H + \beta^L * \overline{X}_g^L + \epsilon_g$. As $\overline{X}_g^H$ and $\overline{X}_g^L$ are perfectly collinear($\overline{X}_g^H + \overline{X}_g^L = 1$), we use $\overline{X}_g^L$ as reference category and we estimate the following regression:

$$\begin{aligned}
\overline{Y_g} &= \beta^H * \overline{X}_g^H + \beta^L * \overline{X}_g^L + \epsilon_g \\
&= \beta^H * \overline{X}_g^H + \beta^L * (1 - \overline{X}_g^H) + \epsilon_g \\
&= \beta^L + (\beta^H - \beta^L) * \overline{X}_g^H + \epsilon_g.
\end{aligned}$$

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	0.213	0.018	12.167	0.000
high_income	0.419	0.032	13.078	0.000

Tab. 2: Naive OLS regression summary

Results are presented in Tab. 2 and show that β^L is estimated to be 0.213, while β^H is estimated to be 0.213+0.419 = 0.632.

That means that the naive OLS regression overestimates the propensity of high-income voters to vote in favour of the reform and underestimates the propensity of low-income voters to vote in favour of the reform, showing a more polarized distribution of votes conditional on income.

That is explained by how OLS attributes higher percentages of votes in favour of the reform: the estimation attributes higher percentages in Northern precincts only to their higher share of high-income voters, while in reality the higher percentages of votes in favour are also driven by the fact that low-income voters in Northern precincts have a higher propensity to vote in favour of the reform than low-income voters in other regions.

4.3.2 `seine` estimates

When we move to semiparametric estimation performed by `seine`, we obtain the results presented in Tab. 3, which are much closer to the true values.

predictor	outcome	estimate	std.error	conf.low	conf.high
high_income	outcome	0.504	0.018	0.469	0.539
low_income	outcome	0.341	0.011	0.319	0.363

Tab. 3: Semiparametric ecological inference estimates

4.3.3 Interaction between $\overline{X}_g$ and Z_g

It can be noted that in this case, the relationship between $E[\beta_g]$ and Z_g is linear.

In fact, if we consider South as the reference category, and we denote with $Z_g = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ the covariate for Northern precincts, with $Z_g = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ the covariate for Central precincts, and with $Z_g = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ the covariate for Southern precincts, we can express η_r as a linear function of Z_g:

$$E[\beta_g|Z_g] = \eta_r = f(Z_g) = \begin{pmatrix} 0.6 \\ 0.2 \end{pmatrix} + \begin{pmatrix} 0 & -0.3 \\ 0.5 & 0.1 \end{pmatrix} Z_g'.$$

Therefore, we can express the model as a linear regression with interaction between $\overline{X}_g$ and Z_g:

$$\begin{aligned} \overline{Y_g} &= f(Z_g) * \overline{X}_g \\ &= (\gamma_0^H + \gamma_1^H * Z^{North} + \gamma_2^H * Z^{Centre}) * \overline{X}_g^H \\ &+ (\gamma_0^L + \gamma_1^L * Z^{North} + \gamma_2^L * Z^{Centre}) * \overline{X}_g^L + \epsilon_g \end{aligned}$$

with γ_0^H representing the propensity of voting in favor of the reform for high-income voters in Southern precincts, γ_1^H representing the difference in propensity for high-income voters between Northern and Southern precincts, γ_2^H representing the difference in propensity for high-income voters between Central and Southern precincts, and analogously for low-income voters with γ_0^L, γ_1^L and γ_2^L: their true values are $\gamma_0^H = 0.6$, $\gamma_1^H = 0$, $\gamma_2^H = -0.3$, $\gamma_0^L = 0.2$, $\gamma_1^L = 0.5$ and $\gamma_2^L = 0.1$.

As $\overline{X}_g^L = 1 - \overline{X}_g^H$, we can rewrite the model as a linear regression with interaction between $\overline{X}_g$ and Z_g and with $\overline{X}_g^H$ as reference category:

$$\overline{Y_g} = \gamma_0^L + (\gamma_0^H - \gamma_0^L) * \overline{X}_g^H + \gamma_1^L * Z^{North} + \gamma_2^L * Z^{Centre} +$$
$$+ (\gamma_1^H - \gamma_1^L) * Z^{North} * \overline{X}_g^H + (\gamma_2^H - \gamma_2^L) * Z^{Centre} * \overline{X}_g^H + \epsilon_g$$

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	0.194	0.008	24.466	0.000
high_income	0.425	0.021	20.210	0.000
area_North	0.504	0.019	26.901	0.000
area_Centre	0.118	0.013	9.002	0.000
high_income:area_North	-0.528	0.032	-16.413	0.000
high_income:area_Centre	-0.437	0.029	-15.033	0.000

Tab. 4: Parametric estimation with covariates regression summary

We can observe that the estimates are very close to the true values, giving us a good approximation of $f(Z_g)$. Through the estimated $f(Z_g)$, we could then obtain estimates of β^H and β^L through previously exposed relationships.

That means that, for each geography, we can compute the relative $\hat{\beta_g^H}$ and $\hat{\beta_g^L}$ as a function of covariates.

In our example, that corresponds to

$$\hat{\beta_g} = \begin{pmatrix} 0.194 + 0.425 \\ 0.194 \end{pmatrix} + \begin{pmatrix} 0.504 - 0.528 & 0.118 - 0.437 \\ 0.504 & 0.118 \end{pmatrix} Z'_g$$

The next passage is to exploit the relationships that link local estimands to global estimands and we get the following global estimates: $\hat{\beta^H} = \frac{\sum_{g=1}^{G} \hat{\beta_g^H} * N_g^H}{\sum_g N_g^H}$ and $\hat{\beta^L} = \frac{\sum_{g=1}^{G} \hat{\beta_g} * N_g^L}{\sum_g N_g^L}$.

In our case, as a result of a quite good estimation of $\hat{\beta}_g$, we get good global estimates, which correspond to the estimates of `seine`: $\hat{\beta^H} = 0.504$ and $\hat{\beta^L} = 0.341$.

beta_high_hat	beta_low_hat
0.504	0.341

Tab. 5: Estimates with covariates

In any case, it is important to note that in order to obtain the same results with a parametric estimation, we had to correctly specify a linear model for $f(Z_g)$.

That is not always possible, and it is relevant that `seine` was able to provide good estimates of β^H and β^L without any previous assumption on the functional form of $f(Z_g)$.

In the Appendix A, I show another example where the relationship between $E[\beta_g]$ and Z_g is non-linear, even if quite simple.

The example presented in Appendix A shows that a simple linear estimator can approximate a non-linear relationship with good precision. Moreover, the results of the estimates obtained with the linear estimator are almost identical to the ones obtained with the `seine` package.

5 An application to Texas electoral data

After having showed the differential performance of `seine` and naive methods in a simple synthetic example of electoral competition, I moved to an application to real data.

In particular, I applied the semiparametric estimation to the 2020 US presidential election in Texas, using the electoral precincts as the geographical units.

5.1 Construction of the dataset

The dataset is constructed by combining electoral results at the electoral precinct level (also referred to as voting districts or VTDs) with demographic data of 2020 US census, with the goal of obtaining, for each precinct, a response variable ($\overline{Y}_g$), a partition of the population in categories ($\overline{X}_g$) and a set of covariates (Z_g).

In the construction of the dataset, I exploited three sources:

- ALARM dataset by T. Kenny and McCartan (2021), which combines electoral results at the electoral precinct level (also referred to as voting districts or VTDs) previously collected by Voting and Election Science Team (2020) with demographic data of 2020 US census collected by U.S. Census Bureau (2020)

- American Community Survey (ACS) by the U.S. Census Bureau, which provide socio-economic characteristics of the population and can be called through the `tidycensus` package in R (Walker and Herman, 2026)

- U.S. Census Bureau (2021)'s files, which provide the mapping of census blocks to CBGs and VTDs, which is necessary to aggregate the socio-economic characteristics collected at the CBG level in ACS surveys to the VTD level.

The construction of the dataset is explained in detail in Appendix B (Section 8).

An important note, however, must be made regarding the quality of the data.

In particular, the ALARM dataset by T. Kenny and McCartan (2021) contains a number of precincts with missing electoral results, even if they have relevant voting age population. At the same time, other precints have a number of votes cast that is higher than the voting age population, which constitutes a clear data error.

In order to overcome these issues, I removed from the dataset all the precincts with less than 10 votes cast (which are either the result of errors or not too relevant for the analysis), and all the precincts with more votes cast than voting age population. Furthermore, I removed also the precincts where the number of votes cast is less than one fifth of the voting age population, as they looked to be outliers in the distribution of the share of votes cast over voting age population.

5.2 Choice of variables and validity of the assumptions

The goal of this application is to estimate the support for the Democratic and Republican candidates in the 2020 US presidential election in Texas among different ethnic groups.

The response variable $\overline{Y}_g$ is the share of votes received by the Democratic candidate in precinct g.

The partition of the population $\overline{X}_g$ is the share of voters in precinct g that belong to each of the four ethnic groups considered in the analysis: White, Black, Hispanic and Other.

Finally, the covariate Z_g is the share of 25 years old or older voters in precinct g that have a college degree, which is collected at the CBG level in ACS surveys and then aggregated to the VTD level as explained in Appendix B (Section 8).

The assumption of bounded N is satisfied by the distribution of the number of voters across precincts being bounded.

The assumption of positivity is satisfied by the distribution of the share of voters in each ethnic group across precincts with varying levels of college degree holders.

Finally, the assumption of CAR cannot be directly tested, but it is driven by the fact that shares of college degree holders in a precinct are likely to capture many of the unobserved characteristics that affect the voting behavior of different ethnic groups in that precinct, such as location of the county (urban vs rural), income, and other socio-economic characteristics.

5.3 Results of the estimation

The results of the estimation are summarized in Table 6, which shows the estimated support for the Democratic (Joe Biden) and Republican (Donald Trump) candidates among each ethnic group, along with the 95% credible intervals.

An evident shortcoming of the results is that the estimates for the support of the candidates among Black voters are not within the [0,1] interval. That, of

predictor	outcome	estimate	std.error	conf.low	conf.high
vap_hisp	pre_20_rep_tru	0.298	0.006	0.287	0.309
vap_white	pre_20_rep_tru	0.858	0.011	0.837	0.878
vap_black	pre_20_rep_tru	-0.083	0.008	-0.097	-0.068
other_ethnicity	pre_20_rep_tru	0.240	0.020	0.201	0.278
vap_hisp	pre_20_dem_bid	0.702	0.009	0.684	0.720
vap_white	pre_20_dem_bid	0.142	0.005	0.133	0.152
vap_black	pre_20_dem_bid	1.083	0.021	1.042	1.123
other_ethnicity	pre_20_dem_bid	0.760	0.021	0.720	0.801

Tab. 6: Semiparametric ecological inference estimates for the 2020 presidential election in Texas

course, should cast doubt on the reliability of the estimates in general, but also on the validity of the methodology, at least with regard to this specific application.

In fact, McCartan and Kuriwaki (2026a) explain that, even if the software `seine` allows to exploit boundedness of response variable in the estimation, it does not guarantee that the estimates will be within the [0,1] interval.

5.4 Results of linear estimation

Previous section and Appendix A have shown that, in our synthetic examples, `seine` package has returned results almost identical to the ones obtained imposing a linear form for $f(Z_g)$.

In order to verify whether that holds also in a true dataset, I have repeated the procedure to estimate β with the following linear specification:

$$\begin{aligned}
\overline{Y_g} &= \gamma_0^{other} * \overline{X}_g \\
&+ (\gamma_0^{hisp} - \gamma_0^{other}) * \overline{X}_g^{hisp} + (\gamma_0^{white} - \gamma_0^{other}) * \overline{X}_g^{white} + (\gamma_0^{black} - \gamma_0^{other}) * \overline{X}_g^{black} \\
&+ \gamma_1^{other} * Z_g * \overline{X}_g^{other} + (\gamma_1^{hisp} - \gamma_1^{other}) * Z_g * \overline{X}_g^{hisp} \\
&+ (\gamma_1^{white} - \gamma_1^{other}) * Z_g * \overline{X}_g^{white} + (\gamma_1^{black} - \gamma_1^{other}) * Z_g * \overline{X}_g^{black} + \epsilon_g
\end{aligned}$$

where Z_g represents the share of people over 25 with a college degree.

Therefore estimates $\hat{\beta}_g$ can be obtained as follows:

$$\hat{\beta}_g = \begin{pmatrix} \hat{\gamma_0}^{other} \\ \hat{\gamma_0}^{hisp} \\ \hat{\gamma_0}^{white} \\ \hat{\gamma_0}^{black} \end{pmatrix} + \begin{pmatrix} \hat{\gamma_1}^{other} \\ \hat{\gamma_1}^{hisp} \\ \hat{\gamma_1}^{white} \\ \hat{\gamma_1}^{black} \end{pmatrix} Z_g.$$

The results are presented in Table 7.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	0.119	0.049	2.405	0.016
vap_hisp	0.199	0.050	3.960	0.000
vap_white	1.118	0.053	21.021	0.000
vap_black	-0.274	0.054	-5.122	0.000
college	0.228	0.078	2.940	0.003
vap_hisp:college	-0.286	0.080	-3.589	0.000
vap_white:college	-1.041	0.086	-12.058	0.000
vap_black:college	-0.039	0.095	-0.408	0.683

Tab. 7: Parametric estimation with covariates for the 2020 presidential election in Texas (Republican share as outcome)

beta_hisp	0.298
beta_white	0.858
beta_black	-0.082
beta_other	0.240

Tab. 8: Parametric group-specific estimates with Texas data

Finally, we can move from local to global estimates, obtaining the results presented in Table 8.

Interestingly enough, the results are almost identical to the ones obtained with `seine`.

6 Conclusions

This thesis has delved into the problem of ecological inference and has shown possible ways to estimate individual relationships from aggregated data without committing the ecological fallacy.

The first important aspect of this thesis is that it has shown that the problem requires a specific notation in order to account for the way in which individuals are aggregated into observation units, which we have called geographies, following Kuriwaki and McCartan (2026), McCartan and Kuriwaki (2026a) and the most obvious applications of ecological inference.

In particular, I have focused on situations in which the population is partitioned into groups, whose shares are collected at geographical level in vectors $\overline{X}_g$, and the researcher is interested in estimating the vector of group-specific propensities β.

Issues in estimation arise from the need to balance two competing desires: on one hand, the desire to allow β to vary across geographies, which is fundamental to give a meaningful interpretation of the data and to avoid the ecological fallacy; on the other hand, the need to impose some structure on the variations of β across geographies in order to allow for a feasible estimation method.

One promising approach, shown by Kuriwaki and McCartan (2026) and McCartan and Kuriwaki (2026a), is to use geography-specific covariates(Z_g) to explain the variations of β across geographies.

In particular, McCartan and Kuriwaki (2026a) show that there is an easily interpretable independence condition, which they call CAR-IND, that is sufficient to write the problem as $E[Y_g] = f(Z_g)'\overline{X_g}$, where Z_g are covariates that explain the variations of β across geographies: $\beta_g = f(Z_g)$.

The interpretation of the CAR-IND condition is that the response variable of an individual is independent of the specific geography in which the individual is aggregated, conditional on the characteristics of the geography, which are represented by the covariates Z_g: that means that the response of an individual does not depend on the specific geography per se, but it depends only on the characteristics of the geography captured by the covariates Z_g.

The role of the researcher, therefore, is to find covariates Z_g that allow for the CAR-IND condition to hold, while, at the same time, they should be independent of the shares $\overline{X}_g$ in order to avoid identification issues.

Once the problem is stated in this way, the methodological issue is to estimate the function $f(Z_g)$.

McCartan and Kuriwaki (2026a) propose a semiparametric estimator that is able to estimate the function $f(Z_g)$ without imposing any functional form, and then to provide estimates of the group-specific propensities. Their methodology is implemented in the `seine` package (McCartan and Kuriwaki (2026b)).

While experimenting with `seine`, I have conducted parallel experiments imposing a linear functional form for $f(Z_g)$, which allowed for a simple OLS model with interactions between the covariates and the shares $\overline{X}_g$.

Estimations with synthetic data have involved using quite simple relationships between covariates and group-specific propensities, experimenting with both linear and non-linear relationships. Both the semiparametric and the linear estimators have provided good estimates of the group-specific propensities, showing effectiveness of the linear approximation of the function $f(Z_g)$, conditioned on the choice of relevant covariates.

When moving to the application to Texas electoral data, estimates from both the semiparametric and the linear estimators have been unsatisfactory, with estimates of the group-specific propensities outside the [0,1] interval. The results were unsatisfactory even when experimenting with different specifications of the model, which I have not reported in this paper.

One possible explanation for the unsatisfactory results is the poor quality of the electoral data in the ALARM dataset, which presents incongruities between the voting age population and the votes cast. Therefore, a possible future direction could be to reconstruct the electoral dataset.

Finally, even when moving to the application to Texas electoral data, the results of the semiparametric estimator have been almost identical to those of the linear estimator. That result is quite surprising, since the Texas electoral data presents a more complex structure, with precincts differing by size and unknown relationships between covariates and group-specific propensities.

That strong resemblance suggests that the semiparametric estimator, as implemented in the `seine` package, tends to return a linear relationship between covariates and group-specific propensities. That would mine the advantage of using a semiparametric estimator, whose advantage is to allow for more flexibility in the search for the functional form of $f(Z_g)$.

In any case, it is worth noting that the methodology is very recent and the `seine` package has been released only a few months ago, so it would be interesting to repeat the experiments with future progress in the methodology and future releases of the software, which could allow for a more flexible search for the functional form of $f(Z_g)$.

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7 Appendix A: A non-linear synthetic example

In this Appendix, I present a synthetic example where the relationship between covariates Z_g and the group-specific propensities β_g is non-linear.

The purpose of this example is to look at the differential performance of `seine`, which is explicitly designed to capture non-linear relationships, and a linear estimator, in estimating non-linear relationships and, finally, the associated group-specific propensities.

7.1 Data generating process

The setup is similar to the one presented in Section 4, with the difference that now we do not have a regional distinction, but rather a continuous covariate Z_g.

Let us suppose that Z_g represents the share of migrants within a precinct, which implies Z_g continuous and $Z_g \in [0, 1]$.

The data generating process is the following.

Precincts differ for their percentage of high-income and low-income voters: the percentages of high-income voters $\overline{X}_g^H$ are extracted from a truncated normal distribution (truncated at 0 and 1) with mean 0.5 and standard deviation 0.2:

$$\overline{X}_g^H \sim \text{TruncNormal}(a = 0, b = 1, \mu = 0.5, \sigma = 0.2).$$

We assume that the share of immigrants Z_g has a loose negative association with the share of high-income voters $\overline{X}_g^H$ (migrants are more likely to live in low-income areas), creating possible identification issues.

We can model this association by assuming that the share of immigrants is extracted from a truncated normal distribution with mean $1-\overline{X}_g^H$ and standard deviation 0.2:

$$Z_g \sim \text{TruncNormal}(a=0, b=1, \mu=1-\overline{X}_g^H, \sigma=0.2).$$

Then, we might assume that

$$\beta_g^H \sim \text{TruncNormal}(a=0, b=1, \mu=\eta_g^H, \sigma=0.05),$$
$$\beta_g^L \sim \text{TruncNormal}(a=0, b=1, \mu=\eta_g^L, \sigma=0.05).$$

The crucial difference with previous setup arises when we set means to be non-linear functions of Z_g:

$$\eta_g^H = 10^{Z_g-1}, \quad \eta_g^L = e^{Z_g-1}.$$

The interpretation of this condition is that both high-income and low-income voters tend to favor the reform more as the share of immigrants in their precinct increases, but the effect is stronger for low-income voters and the relationship is non-linear.

Finally, the observed precinct-level outcome ($\overline{Y}_g$, i.e., the share of voters in favour of the reform in precinct g) is generated as the weighted average of these two group-specific propensities:

$$\overline{Y}_g = \overline{X}_g^H \beta_g^H + \overline{X}_g^L \beta_g^L.$$

7.2 Comparison of results

The true values of the group-specific propensities are $\beta_g^H = 0.324$ and $\beta_g^L = 0.660$.

high	low
0.324	0.660

Tab. 9: True coefficients up to approximation

7.2.1 OLS

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	0.917	0.023	40.423	0.000
high_income	-0.845	0.042	-20.021	0.000

Tab. 10: Naive OLS regression summary

OLS estimates are $\hat{\beta}_g^H = 0.917$ and $\hat{\beta}_g^L = 0.917 - 0.845 = 0.072$.

As we could expect, these estimates are far from the true values as they do not take into account the effect of the covariate Z_g on the group-specific propensities β_g.

7.2.2 `seine`

predictor	outcome	estimate	std.error	conf.low	conf.high
high_income	outcome	0.321	0.014	0.294	0.348
low_income	outcome	0.664	0.023	0.618	0.709

Tab. 11: Semiparametric ecological inference estimates

As in the example presented in Section 4, `seine` is able to provide good estimates of the group-specific propensities.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	0.176	0.022	7.866	0.000
high_income	-0.082	0.034	-2.403	0.017
zeta	0.869	0.033	26.049	0.000
high_income:zeta	-0.357	0.060	-5.961	0.000

Tab. 12: Parametric estimation with covariates regression summary

7.2.3 Interaction between $\overline{X}_g$ and Z_g

In this case, the functional form of the relationship between Z_g and β_g is estimated to be linear, which is not the case in the data generating process. In particular, it is estimated that $E[\beta_g^L|Z_g] = \gamma_0^L + \gamma_1^L * Z_g$, while $E[\beta_g^H|Z_g] = \gamma_0^H + \gamma_1^H * Z_g$.

The estimates are $\hat{\beta}_g^L = 0.176 + (0.869) * Z_g$ and $\hat{\beta}_g^H = (0.176 - 0.082) + (0.869 - 0.357) * Z_g$.

Then, we can adopt the same strategy presented in Section 4 to produce global estimates.

If we plug in the observed values of Z_g and $\overline{X_g}$ in the above equations, we obtain estimates of β_g for each precinct g. The weighted average of these estimates across precincts is $\hat{\beta}^H = 0.321$ and $\hat{\beta}^L = 0.664$, which are in fact quite close to the true values of β^H and β^L and correspond to estimates performed with `seine`.

beta_high_hat	beta_low_hat
0.321	0.664

Tab. 13: Estimates with covariates

The conclusion that we can draw from this example is that, even if the relationship between Z_g and β_g is non-linear, a linear approximation of this relationship can still provide good estimates of the group-specific propensities β_g, as far as the relevant covariates are used.

8 Appendix B: Construction of the dataset

The dataset to perform the analysis should include, at the most granular level possible, the share of votes received by the candidate ($\overline{Y}_g$), a partition of the voting population in categories of interest (e.g, ethnic groups, which represent $\overline{X}_g$) and covariates at that same geographical level (Z_g).

The most granular level for electoral results in the United States is the electoral precinct, whose dimension is typically in the order of hundreds or a couple of thousands of voters.

Census data, instead, are collected at different geographical levels, with geographical levels varying with the variables of interest.

The most granular level for census data is the census block, which is a geographical unit that is typically bounded by geographical features either natural (e.g., rivers) or artificial (e.g., streets). Given their geographical definition, census blocks are quite different in population size, with a considerable amount of blocks being uninhabited, and the most populated ones (typically in urban areas) having a population of hundreds of people.

The only variables collected at the census block level are the total population, the voting age population, ethnic composition and the number of housing units. Moreover, these variables are collected at the census block level only in the decennial census, which is conducted every ten years.

Census blocks are then grouped into larger geographical units, which are used to collect other variables of interest. In particular, the census block groups (CBGs) are the smallest geographical unit for which the American Community Survey (ACS) collects data on a yearly basis. CBGs are typically composed of a few hundred to a few thousand people, and they are designed to be relatively homogeneous in terms of population characteristics.

Variables collected at the CBG level in ACS include several socio-economic characteristics, such as income, educational attainment, employment status, as well as language spoken at home, commuting patterns, and housing characteristics. The ACS data are collected through sample surveys, so that data at the CBG level are subject to sampling variability.

Finally, census block groups are grouped into larger geographical units, which are called census tracts.

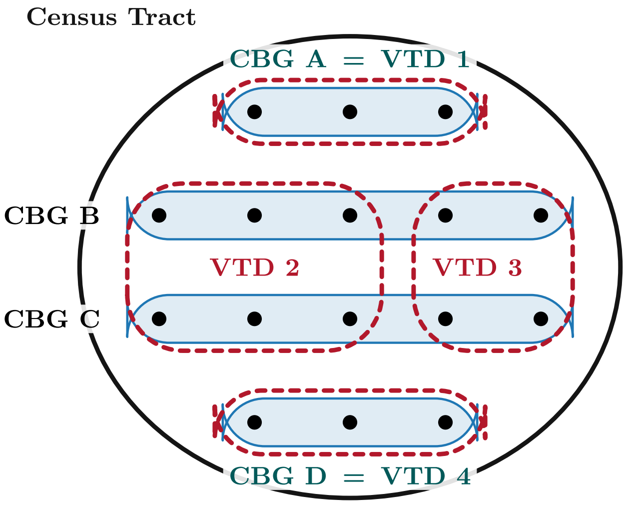
The electoral precincts, instead, are geographical units that are defined by the electoral authorities for the purpose of organizing elections. Precincts are typically composed of a few hundred to a few thousand voters, and they are sometimes denoted as voting districts (VTDs).

VTDs, however, are not units of the census, and they are not completely aligned with CBGs. VTDs are always constituted by a set of census blocks, and they are contained within a single census tract, but they can cross the boundaries of CBGs. This means that a VTD can contain parts of multiple CBGs, and a CBG can be split across multiple VTDs; viceversa is also true.

Fig. 1 shows a schematic representation of a possible relationship between census blocks, CBGs, census tracts and VTDs.

Some of the variables of possible interest for the analysis are collected only at the CBG level in ACS surveys. Therefore, a procedure to aggregate the data at the VTD level was necessary.

ALARM dataset by T. Kenny and McCartan (2021) combine electoral results at the VTD level previously collected by Voting and Election Science Team (2020) with demographic data of 2020 US census collected by U.S. Census



● Census block   CBGs   VTDs

Census blocks are the atomic units. CBGs and VTDs are two independent partitions of the same census tract, using census blocks as atomic units. VTD 1 and VTD 4 coincide exactly with CBG A and CBG D, while VTD 2 and VTD 3 cut across CBG B and CBG C. Both partitions are entirely contained within the census tract.

Fig. 1: Relationship between electoral precincts (VTDs), census blocks, census block groups (CBGs), and census tracts.

Bureau (2020).

That provides a dataset that includes the share of votes received by the candidates at the VTD level, as well as the total population and the voting age population at the census block level with the corresponding ethnic composition.

This dataset, while being very useful, does not include the socio-economic characteristics, which are of fundamental importance for the analysis, and which are collected at the CBG level in ACS surveys. ACS data are easily accessible through the `tidycensus` package in R (Walker and Herman, 2026).

The following passage involved exploiting U.S. Census Bureau (2021)'s files, which provide the mapping of census blocks to CBGs and VTDs, to aggregate the socio-economic characteristics collected at the CBG level in ACS surveys to the VTD level.

As in Fig. 1, we can think of three scenarios: VTDs that perfectly align with one CBG, VTDs that align with exactly two or more CBGs, and VTDs that cross the boundaries of CBGs.

For the first scenario, the aggregation of socio-economic characteristics from CBGs to VTDs is straightforward: the characteristics of the VTD are simply the same as those of the corresponding CBG.

The second scenario is also straightforward: the socio-economic characteristics of the VTD are simply the either the sum (in case of counts, e.g, population or the number of people with a college degree) or the average (in case of other quantities, e.g. median income) of the socio-economic characteristics of the CBGs that compose it.

The third scenario is a bit more complicated and requires a procedure to apportion the socio-economic characteristics of the CBGs that are crossed by the VTD boundaries.

The procedure rests on the assumption that differences in the apportionment of census blocks in the construction of the VTDs and CBGs are independent of the socio-economic characteristics of the population.

That assumption is partly disputable, because the electoral authorities in the United States have been known to engage in gerrymandering, which is the practice of drawing electoral district boundaries in a way that favors a particular political party or group. However, gerrymandering is typically done at the level of congressional districts, without particular interventions at the level of VTDs.

Moreover, census block groups are designed to be relatively homogeneous in terms of population characteristics, so that the differences in apportionment of census blocks between VTDs and CBGs are likely to be small.

Therefore, I have assumed that the average socio-economic characteristics of the population in a CBG are the same as those of the population in the census blocks that compose it. Subsequently, I have constructed fictitious populations of the census blocks, assigning to each census block the average socio-economic characteristics of the population in the corresponding CBG.

I will give two examples of the procedure, one for a count variable and one for a non-count variable: if a CBG with population of 1000 people has 200 people with a college degree, I have assumed that each census block in that CBG has 20% of its population with a college degree, so that a census block with 100 people has 20 people with a college degree; on the other side, if a CBG has a median income of $50,000, I have assumed that each census block in that CBG has a median income of $50,000.

Finally, I have aggregated the fictitious populations of the census blocks to the VTD level, obtaining an estimate of the socio-economic characteristics of the population in each VTD. Obviously, these estimates are subject to some error, partly arising from the sampling errors in the original ACS data, and partly from the apportionment procedure.

This procedure is easily implementable and can be applied to any variable of interest that is collected at the CBG level.

9 Code

All the code is available in the GitHub repository, at the following URL: `https://github.com/GiovanniStivella/ecological-inference-master-thesis/`.

In particular, the `Data` folder contains the code to construct the dataset, while the `Code` folder contains the code to perform the estimation and generate the figures and tables of this paper. Through the scripts contained in the `Code` folder, it is possible not only to replicate the results of this paper, but also to try different specifications of the model, by changing the covariates included in the estimation. In fact, R scripts contain also other experiments that I have chosen not to include in the paper.

Finally, the `Paper` folder contains the code to generate this paper, which is written in LaTeX.